

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

600 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

600 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.17 Cr after input tax credit, generating 9,19,800 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹4.39 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	600 kWp
Year 1 generation	9,19,800 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹65.86 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.28 Cr
GST at 5% — recovered as ITC	₹10.86 L
Net capitalised cost	₹2.17 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹54.65 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,17,14,285	86,85,714	1,30,28,571	21,86,020
2	1,30,28,572	52,11,429	78,17,143	13,11,612
3	78,17,143	31,26,857	46,90,286	7,86,967
4	46,90,285	18,76,114	28,14,171	4,72,180
5	28,14,172	11,25,669	16,88,503	2,83,308
6	16,88,503	6,75,401	10,13,102	1,69,985
7	10,13,102	4,05,241	6,07,861	1,01,991
8	6,07,861	2,43,144	3,64,717	61,194
9	3,64,717	1,45,887	2,18,830	36,717
10	2,18,830	87,532	1,31,298	22,030
11	1,31,298	52,519	78,779	13,218
12	78,779	31,512	47,267	7,931
Tax shield, years 1-12				54,53,155

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

600 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	9,19,800	65,85,768	—	86,85,714	(5,28,514)	71,14,282	71,14,282
2	9,15,201	68,14,953	3,60,000	52,11,429	3,12,970	61,41,983	1,32,56,265
3	9,10,625	70,52,113	3,60,000	31,26,857	8,97,304	57,94,809	1,90,51,074
4	9,06,072	72,97,527	3,60,000	18,76,114	12,73,856	56,63,670	2,47,14,744
5	9,01,542	75,51,481	3,60,000	11,25,669	15,26,644	56,64,837	3,03,79,581
6	8,97,034	78,14,272	3,60,000	6,75,401	17,06,106	57,48,166	3,61,27,747
7	8,92,549	80,86,209	3,60,000	4,05,241	18,42,541	58,83,667	4,20,11,414
8	8,88,086	83,67,609	3,60,000	2,43,144	19,54,160	60,53,448	4,80,64,862
9	8,83,645	86,58,802	3,60,000	1,45,887	20,51,926	62,46,876	5,43,11,738
10	8,79,227	89,60,128	3,60,000	87,532	21,42,450	64,57,678	6,07,69,416
11	8,74,831	92,71,940	3,60,000	52,519	22,29,739	66,82,201	6,74,51,617
12	8,70,457	95,94,604	3,60,000	31,512	23,16,234	69,18,370	7,43,69,987
13	8,66,105	99,28,496	3,60,000	18,907	24,03,441	71,65,055	8,15,35,042
14	8,61,774	1,02,74,008	3,60,000	11,344	24,92,302	74,21,705	8,89,56,747
15	8,57,465	1,06,31,543	3,60,000	6,806	25,83,429	76,88,114	9,66,44,861
16	8,53,178	1,10,01,521	3,60,000	4,084	26,77,230	79,64,291	10,46,09,152
17	8,48,912	1,13,84,374	3,60,000	2,450	27,73,998	82,50,376	11,28,59,528
18	8,44,667	1,17,80,550	3,60,000	1,470	28,73,954	85,46,596	12,14,06,124
19	8,40,444	1,21,90,513	3,60,000	882	29,77,282	88,53,232	13,02,59,356
20	8,36,242	1,26,14,743	3,60,000	529	30,84,141	91,70,603	13,94,29,959
21	8,32,061	1,30,53,736	3,60,000	318	31,94,680	94,99,057	14,89,29,016
22	8,27,900	1,35,08,006	3,60,000	191	33,09,042	98,38,964	15,87,67,980
23	8,23,761	1,39,78,085	3,60,000	114	34,27,371	1,01,90,714	16,89,58,694
24	8,19,642	1,44,64,522	3,60,000	69	35,49,809	1,05,54,713	17,95,13,407
25	8,15,544	1,49,67,887	3,60,000	41	36,76,503	1,09,31,385	19,04,44,792
26	8,11,466	1,54,88,770	3,60,000	25	38,07,603	1,13,21,167	20,17,65,959
27	8,07,409	1,60,27,779	3,60,000	15	39,43,263	1,17,24,516	21,34,90,475
28	8,03,372	1,65,85,546	3,60,000	9	40,83,643	1,21,41,903	22,56,32,378
29	7,99,355	1,71,62,723	3,60,000	5	42,28,908	1,25,73,815	23,82,06,193
30	7,95,358	1,77,59,986	3,60,000	3	43,79,228	1,30,20,758	25,12,26,951
Total	2,56,83,724	33,88,58,194	1,04,40,000	2,17,14,281	7,71,91,243	25,12,26,951	25,12,26,951

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

600 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹4.39 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹25.12 Cr
30-year generation	2,56,83,724 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,56,83,724 units, avoiding 21,060.7 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.28 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 600 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.